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Circularly-polarized antenna resonates at 1.575GHz.

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Abstract

The present project aims to design a right-handed circularly polarized (CP) antenna which resonates at 1.575 GHz. The utilized software is HFSS. The use of this kind of device is especially relevant in GNSS such as GPS or BDS-3, which happens to be my final project research area. The main requirements to be fulfilled are a bandwidth over 100 MHz, a return loss lower than -10 dB and an axis ratio no higher than 2 dB. The design of the antenna and its behavior simulation is carried out with HFSS software. At the beginning, some innovative designs based on microstrip technology are tested from research: the first one uses two perpendicular monopoles in a S-like shape, which together with a Wilkinson power divider gives a circular polarization; the second one is a more advanced implementation, which uses two floors of substrate and up to three strips with very specific geometry to get the same result. However, given the failure of their implementation for unknown reasons, a conventional approach like the one explained in class is finally taken as a last resource to fulfill the requirements.

Key words: GNSS, RHCP, Microstrip, Wilkinson.

1 Introduction

1.1 Background

Global Navigation Satellite Systems (GNSS) are matrices of satellites orbiting Earth that provide us with an accurate Position, Navigation and Timing (PNT) data that can later be processed to estimate one's localization in real-time. The most popular example is the Global Positioning System (GPS), developed by the U.S. Department of Defense in the 90s, but the following decades many others have emerged to sustain the interests of the different powers in the world: GLONASS in Russia, Galileo in Europe and Beidou System (1, 2 and 3) in China, being this last one present in many of Chinese allied countries (e.g. South Africa).

Regarding details about GNSS antennas, the L1 band is usually chosen because of its low attenuation and its compatibility, and thus its frequency of 1.575 GHz is adopted. Additionally, Right Hand Circular Polarization (RHCP) is preferred in GNSS communications because it distinguishes better signal reflections and multipath conditions compared to linear polarization. Moreover, GNSS antennas require a relatively narrow bandwidth to maintain consistent reception across challenges.

Given the fact that my final research project is focused on GNSS signal processing (I am currently investigating spoofing attacks and anti-spoofing strategies), this kind of antenna seems rather interesting for me for its possible future applications. For instance, an anti-spoofing technique called Angle-of-Arrival (AOA) or Direction-of-Arrival (DOA) consists on detecting the origin of the signal transmitted to detect if it is an expected signal or an attack from a spoofer (malicious device with the intention of misguiding the receiver). **This particular technique has a lot to do with the physical layer and concretely with the antenna's role in reception and familiarizing myself with this one seems a logical step in my research.** Thus, I chose the circularly polarized antenna for my project.

1.2 Research Status

Straightforward methodology

In [1] a conventional design for a right-handed circularly polarized antenna is given, which is applied for GPS. The main idea is to get two orthogonal EM fields which will produce a uniform radiation around the antenna. While directional orthogonality is achieved through the inclination of two separate meandered monopoles, an additional Wilkinson power divider is located between the feed and these monopoles to get a perpendicular phase as well. This Wilkinson device basically consists of the path separation of the microstrip and a resistor in between them. Finally, a defected ground structure (DGS) is incorporated to improve isolation.

Area-constrained alternative

Paper [2] also describes a microstrip antenna but gives a more advanced alternative for the same requirements but additional space constraints (i.e. it is smaller). In this case, there are two layers of substrate and up to three strip designs. The antenna uses a slotted ground and a radiation patch fed by a T-like coupling feedline.

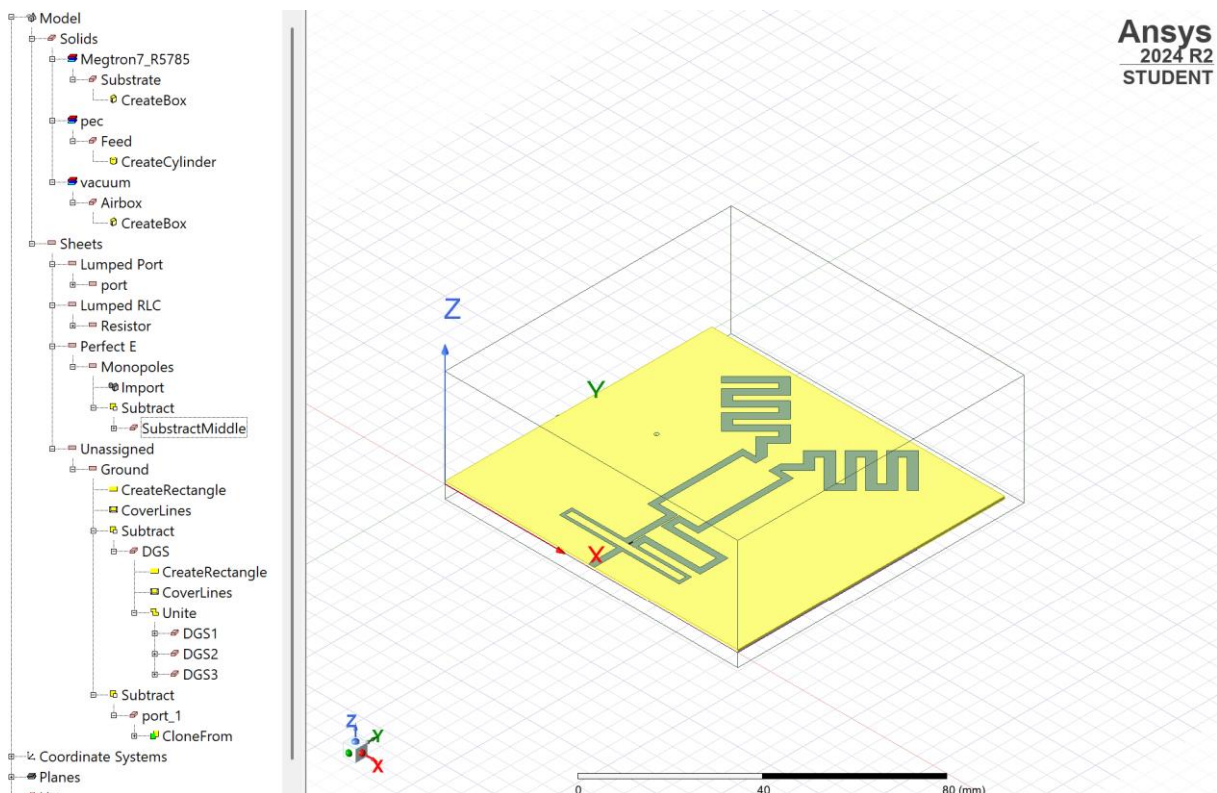
2 Antenna design

The geometry deconstruction for each of the microstrips is based on the values provided by each paper.

Note: while saving the straightforward design, the union of the microstrip lost all the primitive rectangles that formed the final design.

Straightforward Design

This microstrip is mainly composed of three parts: the feed, the Wilkinson divider and the two monopoles. All of them are basically made of rectangles that are united and subtracted, although the last get inclined as well. The geometry of the following basic rectangles is the following:



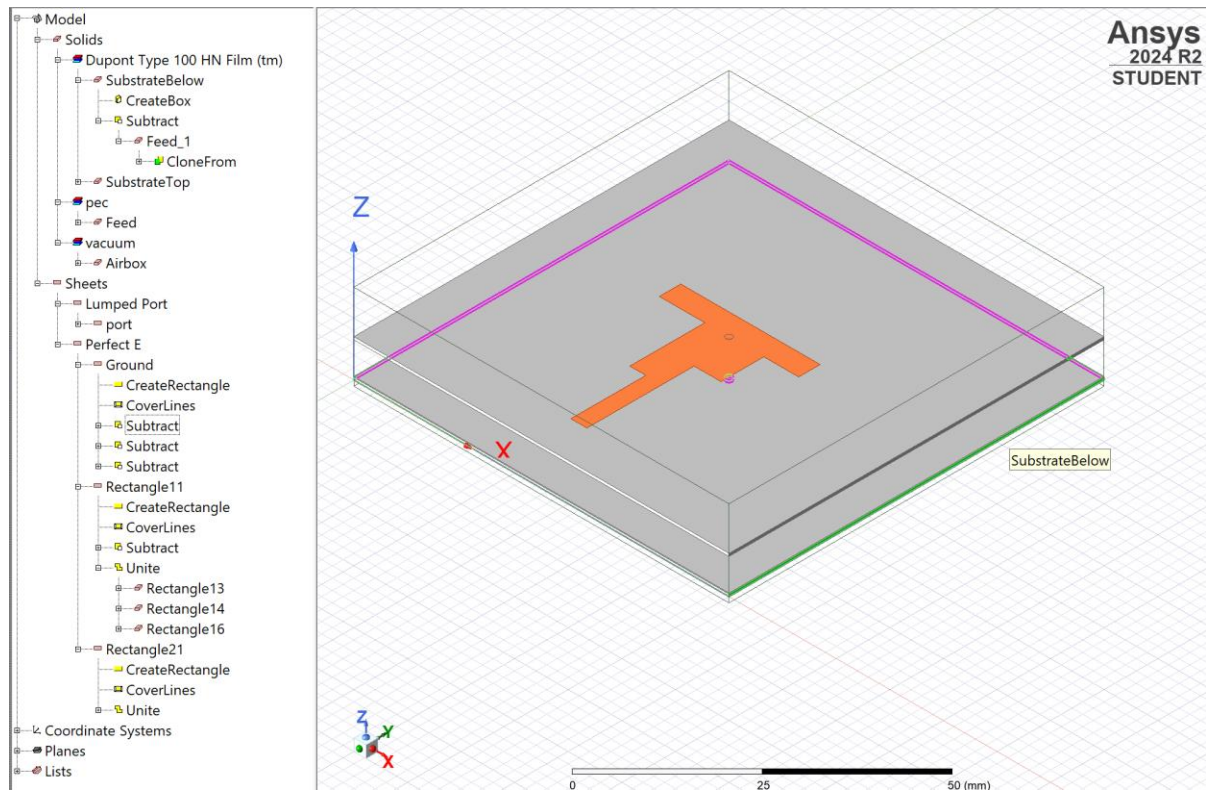
- Rectangle 1
 - Height = ? (we set $L3=8.6$ because it SEEMS that length)
 - Width = $W1 = 1.8$
 - Position = $(44.1, 0)$
- Rectangle 2
 - Height = $g1 + 2*W2 = 3.8$
 - Width = $W1 + 2*L1 = 36.8$
 - Position = $(26.6, 8.6?)$
 - Subtract
 - Height2 = $g1 = 2$
 - Width2 = $36.8 - 2*W2 = 35$
 - Position = $(26.6 + W2, 8.6 + W2) = (27.5, 9.5)$

- Rectangles 3
 - Height = $L2 = 12.6$
 - Width = $W3 = 1.8$
 - $Pos1 = (45 - 0.2 - W3, L3? + g1 + 2*W2) = (43, 12.4)$ [Chosen the space between lines to be 0.4, half from the center is 0.2]
 - $Pos2 = (45.2, 12.4)$
- Rectangle 4
 - Height = $L3 = 8.6$
 - Width = $L4 + W3 - W3 = 17$
 - $Pos = (45 + 0.2 + W3, L3? + g1 + 2*W2 + (L2 - L3)/2) = (47, 14.4)$
 - Subtract (also from Rect3)
 - Height = $L3 - 2*W3 = 5$
 - Width = $L4 = 17$
 - $Pos = (45.2, L3? + g1 + 2*W2 + (L2 - L3)/2 + W3) = (45.2, 16.2)$
- Rectangles 5
 - Height = $W3 = 1.8$
 - Width = $L6 + 2*W3 = 7.6$
 - $Pos1 = (43 - W3 - L6, 12.4 + L2) = (37.2, 25)$
 - $Pos2 = (45.2, 25)$
- Rectangles 6
 - Height = $L5 = 25.7$
 - Width = $W3 = 1.8$
 - $Pos1 = (37.2, 12.4 + L2 + W3) = (37.2, 26.8)$
 - $Pos2 = (45.2 + W3 + L6, 12.4 + L2 + W3) = (51, 26.8)$
- Rectangles 7
 - Height = $W3 = 1.8$
 - Width = $L6 + W3 = 5.8$
 - $Pos1 = (37.2, 12.4 + L2 + W3 + L5) = (37.2, 52.5)$
 - $Pos2 = (45.2 + W3, 12.4 + L2 + W3 + L5) = (47, 52.5)$
- Rectangles 8
 - Height = $W3 = 1.8$
 - Width = $W3 = 1.8$
 - $Pos1 = (37.2 + L6, 52.5 + W3) = (41.2, 54.3)$
 - $Pos2 = (47, 52.5 + W3) = (47, 54.3)$
- Base 1
 - Height = $L7 = 5$
 - Width = $W4 = 2.5$
 - $Pos = (41.2, 54.3 + 1.8) = (41.2, 56.1)$
- Base 2
 - Height = $W4 = 2.5$
 - Width = $L7 = 5$
 - $Pos = (47, 54.3 + 1.8) = (47, 56.1)$
- Base 11
 - Height = $L8/2 = 7.5$

- Width = $W4 = 2.5$
 - Pos = $(41.2, 56.1 + L7) = (41.2, 61.1)$
- Base 22
 - Height = $W4 = 2.5$
 - Width = $L8/2 = 7.5$
 - Pos = $(47 + L7, 56.1) = (52, 56.1)$
- S1
 - Height = $5*W4 + 5*g3 = 12.5 + 10 = 22.5$
 - Width = $L8 = 15$
 - Pos = $(41.2 - L8/2, 61.1 + W4) = (33.7, 63.6)$
 - Subtract1i
 - Height = 2.5
 - Width = 12.5
 - Pos = $(33.7, 63.6 + 4.5*(i-1))$ for $i = 1, 3, 5$
 - Pos = $(33.7 + 2.5, 63.6 + 4.5*(i-1))$ for $i = 2, 4$
- S2
 - Height = $L8 = 15$
 - Width = $5*W4 + 5*g3 = 12.5 + 10 = 22.5$
 - Pos = $(52 + W4, 56.1 - L8/2) = (54.5, 48.6)$
 - Subtract2i
 - Height = 12.5
 - Width = 2.5
 - Pos = $(54.5 + 4.5*(i-1), 48.6)$ for $i = 1, 3, 5$
 - Pos = $(54.5 + 4.5*(i-1), 48.6 + 2.5*(i-1))$ for $i = 2, 4$
- DGS central
 - Height = $L9 = 24$
 - Width = $g4 = 0.8$
 - Pos = $(45 - g4/2, Lg - L9) = (44.6, 31)$
- DGS below
 - Height = $g4 = 0.8$
 - Width = $L11 = 6.6$
 - Pos = $(45 - L11/2, 31 - g4) = (41.7, 30.2)$
- DHS lateral
 - Height = $L10 = 10$
 - Width = $g4 = 0.8$
 - Pos1 = $(45 - L11/2, 31) = (41.7, 31)$
 - Pos2 = $(45 + L11/2 - g4, 31) = (47.5, 31)$

Space-constrained Design

This microstrip is also composed of basic rectangles, although there are several layers with different heights. A particularity is the slotted ground plane, whose shape includes inclined forms as well. The geometry of the following basic rectangles is the following:



Ground plane

- Ground
 - Height = $G1$
 - Width = $G1$
 - Position = $(0, 0, 0)$
 - Substract - Rectangle1
 - Height = $2 \cdot G4$
 - Width = $2 \cdot G4$
 - Position = $(G1/2 - G4, G1/2 - G4)$
 - Substract - Rectangle2
 - Height = $2 \cdot G5$
 - Width = $2 \cdot G5$
 - $(G1/2 - G5, G1/2 - G5)$
 - Substract - Rectangle3467
 - Height = $dcut \cdot \sqrt{2}$
 - Width = $dcut \cdot \sqrt{2}$

- Inclination = 45 degrees
- Position: move their centers to the corners of Rectangle1
- Subtract – Rectangles58910
 - Height = $d_{cut} \cdot \sqrt{2}$
 - Width = $d_{cut} \cdot \sqrt{2}$
 - Inclination = 45 degrees
 - Position: align on the inside with borders of Rectangles3467

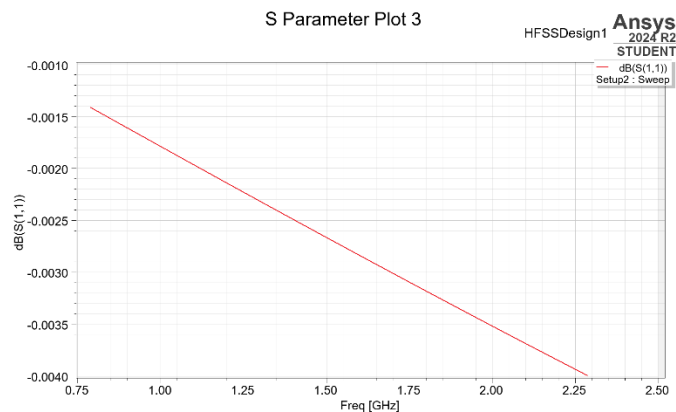
Blue plane

- Rectangle11
 - Height = G_2
 - Width = G_2
 - Position = $(G_1/2 - G_2/2, G_1/2 - G_2/2)$
 - Subtract - Rectangle 12
 - Height = G_3
 - Width = G_3
 - Position = $(G_1/2 - G_3/2, G_1/2 - G_3/2)$
- Rectangle13
 - Height = W_3
 - Width = L_3
 - Position = $(G_1/2 + G_3/2 - L_3, G_1/2 - G_3/2 + W_3)$
- Rectangle14
 - Height = $L_1 + W_t$
 - Width = $W_1 + W_t$
 - Position = $(G_1/2 - G_3/2, G_1/2 + G_3/2 - L_1 - W_t)$
 - Subtract – Rectangle15
 - Height = L_1
 - Width = W_1
 - Position = $(G_1/2 - G_3/2, G_1/2 + G_3/2 - L_1)$
- Rectangle16
 - Height = $W_2 + W_t$
 - Width = $L_2 + W_t$
 - Position = $(G_1/2 - G_3/2, ?)$
 - Subtract – Rectangle17
 - Height = W_2
 - Width = L_2
 - Position = $(G_1/2 - G_3/2, ?)$

3 Results

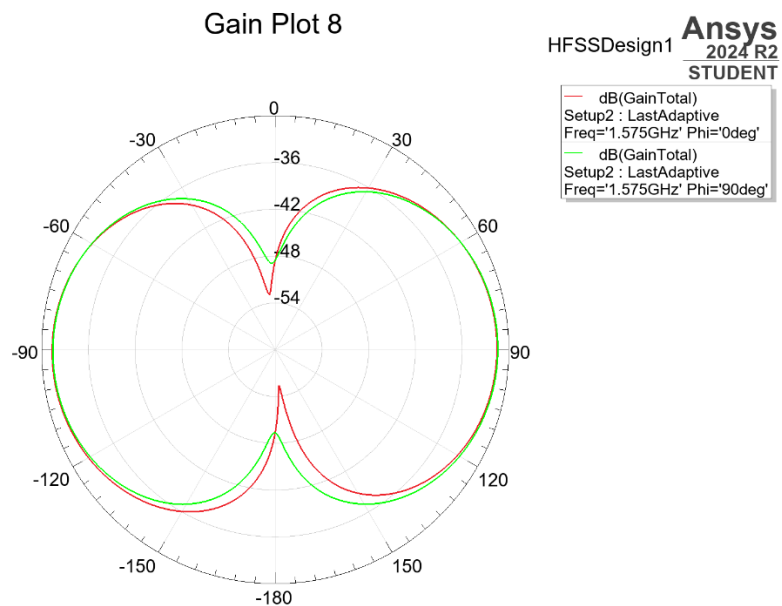
3.1 Straightforward

S11



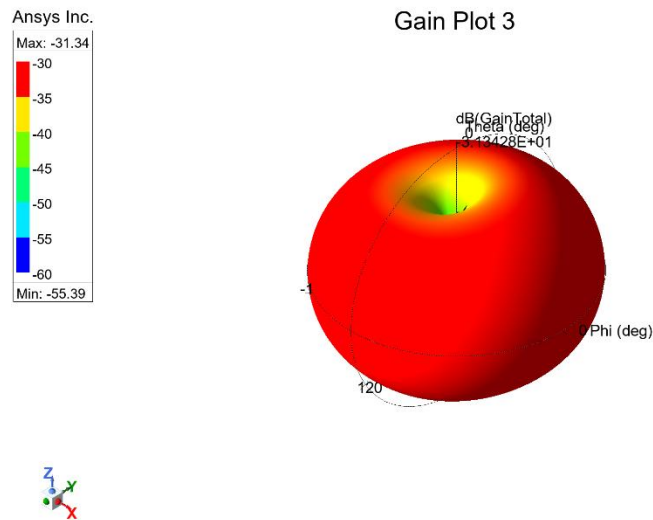
Taking a look at the scale, the return loss can be rounded to zero. This is the main indicator of a wrong design and every next result is one way or another related to it.

Radiation pattern



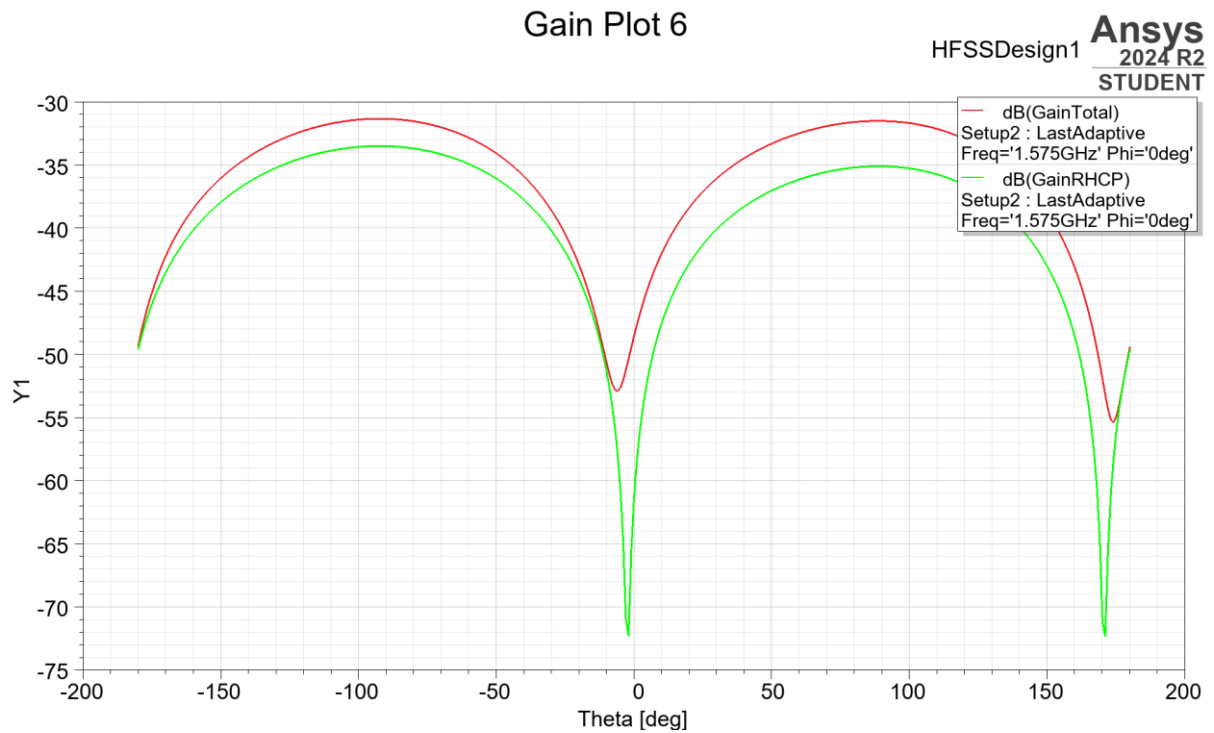
Although the polar plots of both planes are pretty aligned and the shape seem somehow in a “middle way” to being correct, this cannot be considered a circular polarization satisfying the requirements.

3D Polar plot



As the 3D plot shows, the result is not a sphere but rather a toroid, which gives a bittersweet feeling of being in a proper direction but with some details to be fixed.

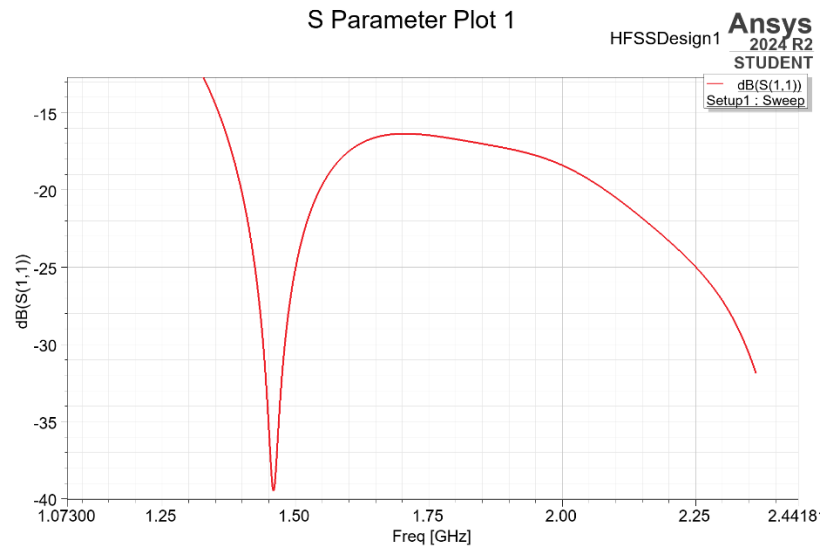
Rectangular gain



The gain graph shows the maximum gain is absolutely absurd, which is directly related to the null return loss shown before.

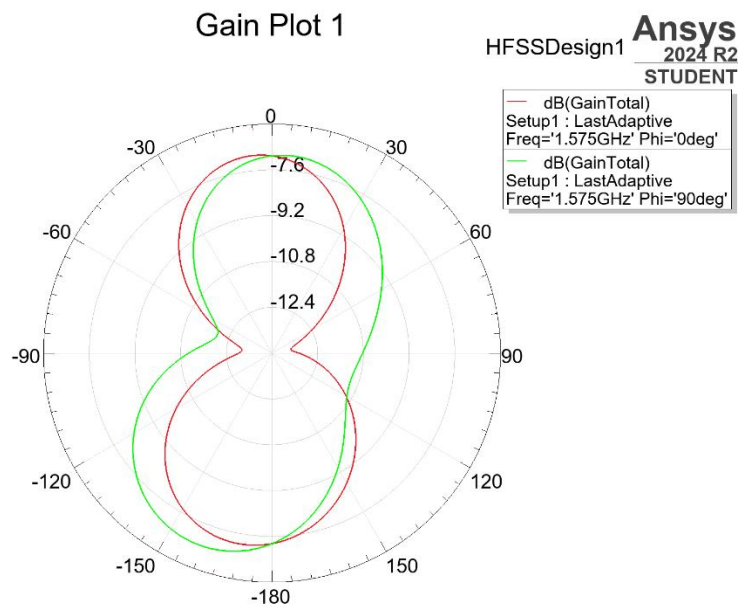
3.2 Space-friendly

S11



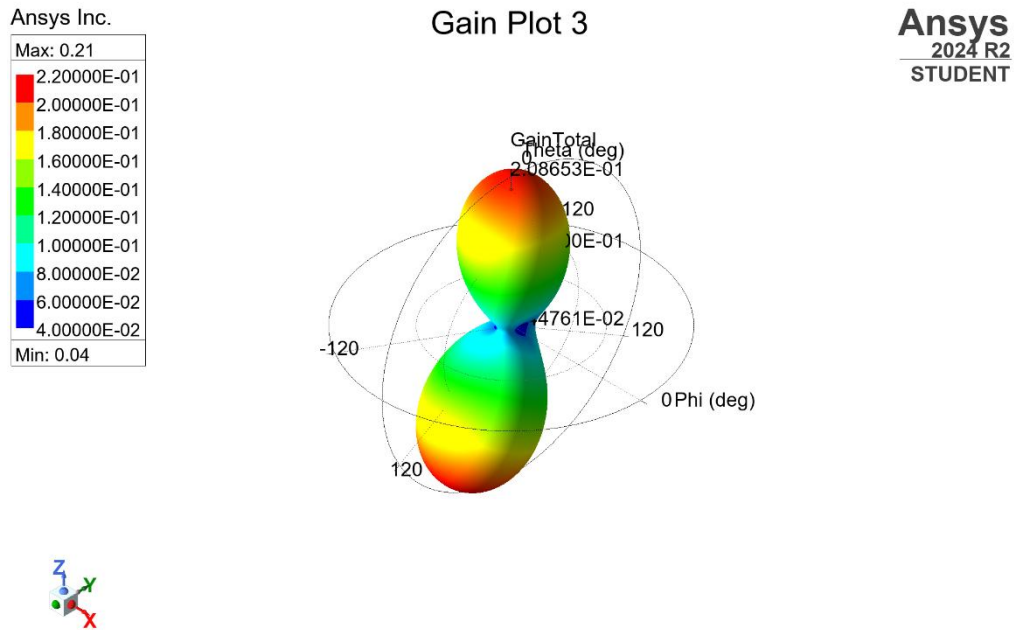
Gladly enough, this second approach does satisfy the return loss requirement by a mile. All frequencies over 1.40 GHz have a return loss below -15 dB, being the lowest almost -40 dB.

Radiation pattern



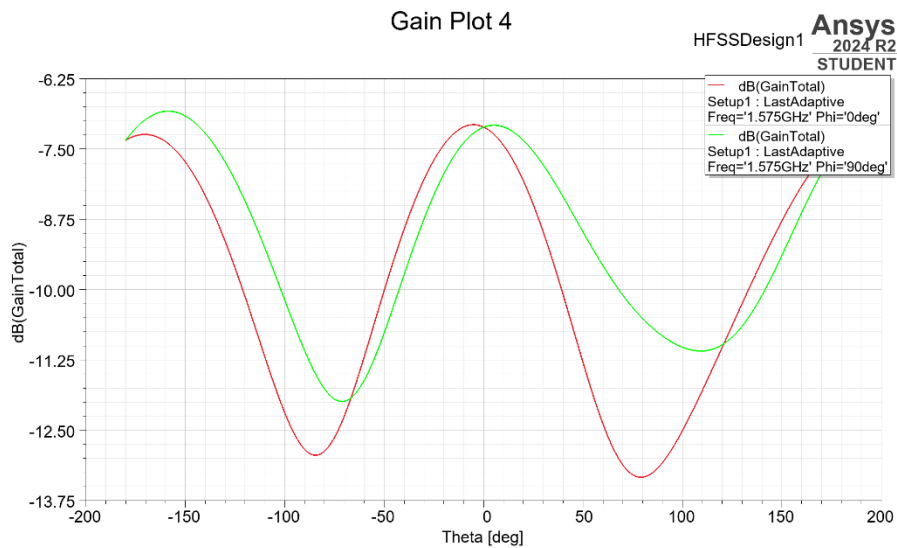
On the other hand, the radiation pattern tends to a more directional antenna with two defined lobes. Besides, the fields alignment is pretty poor, whose consequences are appreciated in the following graph.

3D Polar plot

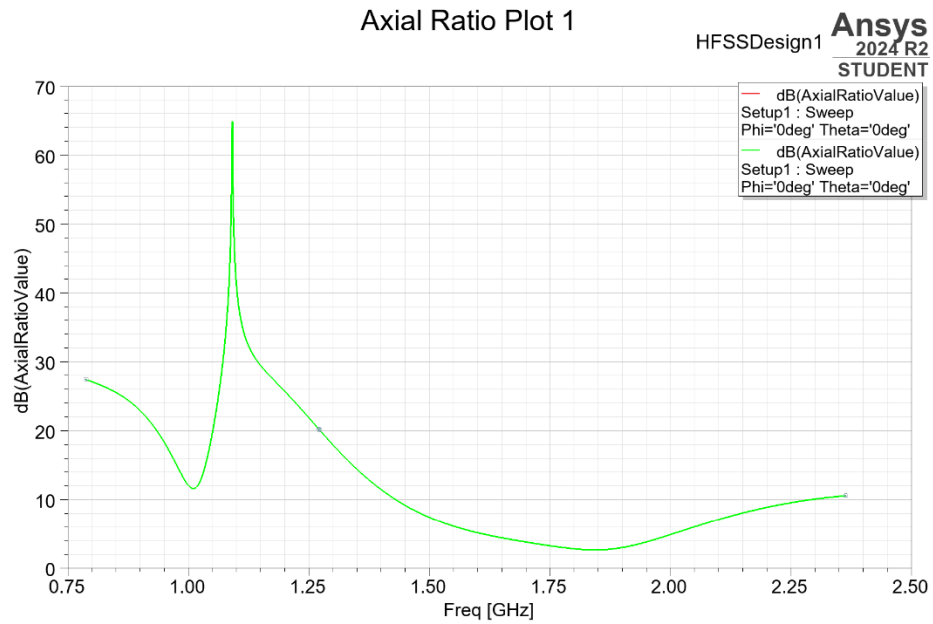


As described before, the gain is far from the desired circular polarization. This might be related to the inactivity of one of the layers, since the one below is not fed and may need an additional coaxial feed line.

Rectangular gain



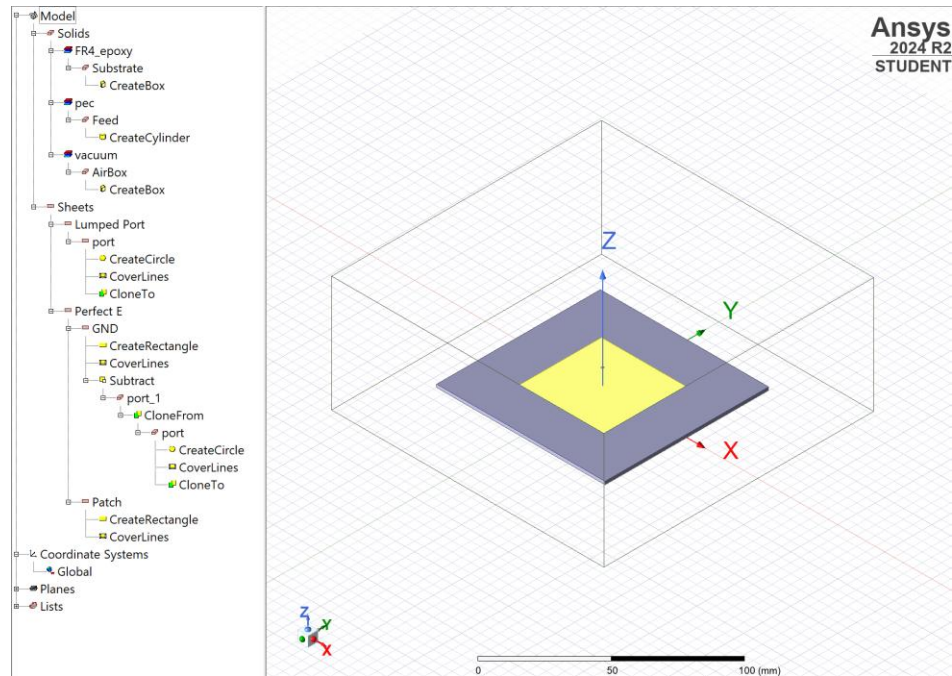
This time, the gain is four times better than the previous approach but is still improvable.



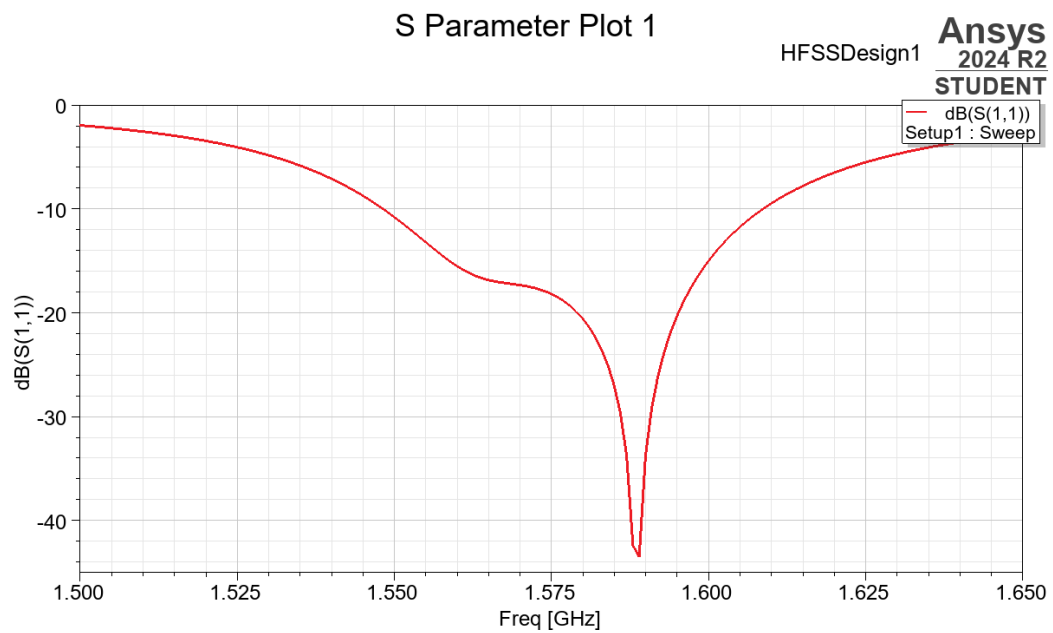
Axial ratio never goes below 2dB and at the central frequency goes well over it.

3.3 Square patch

Similar to the simulation, the results this time do satisfy the stated requirements for a GNSS antenna.

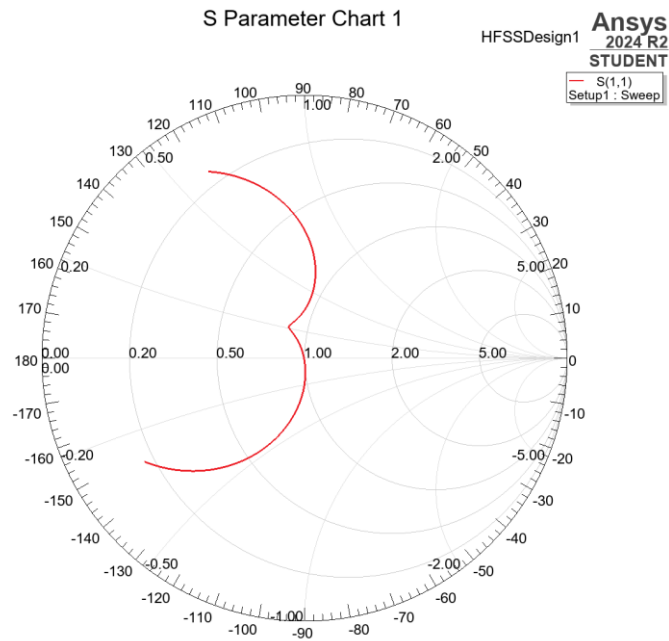


S11

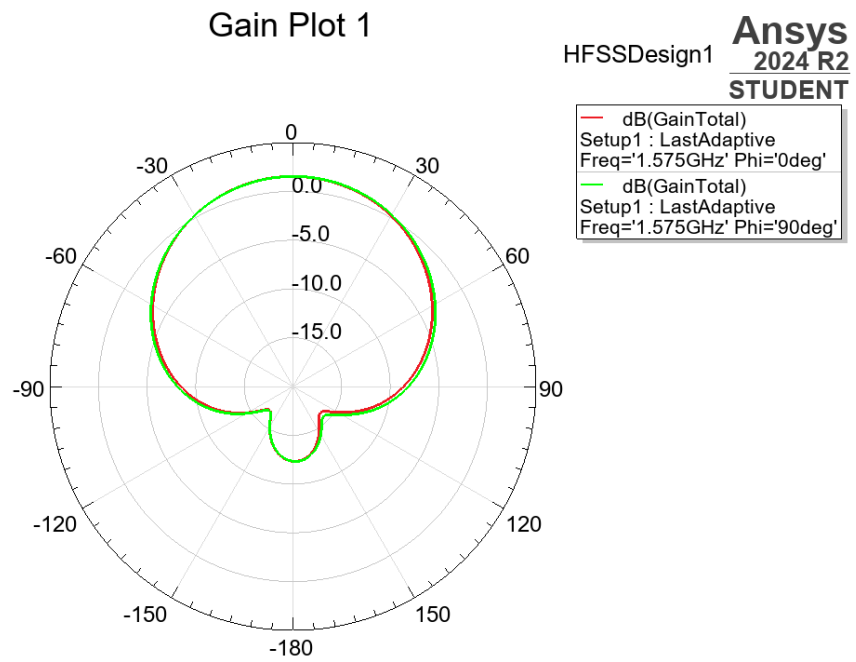


The return loss is lower than -10 dB for a wide range of the frequency, with the lowest of all around 1.59GHz.

Smith Chart



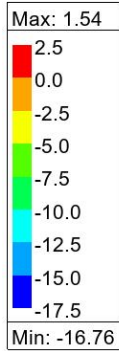
Radiation pattern



The polar plot shows almost a perfect alignment between E and H planes, giving very promising results for the 3D plot circular polarization.

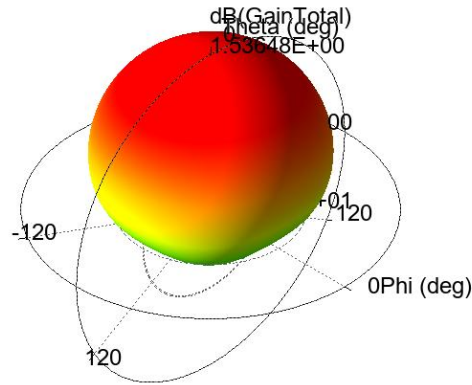
3D Polar plot

Ansys Inc.



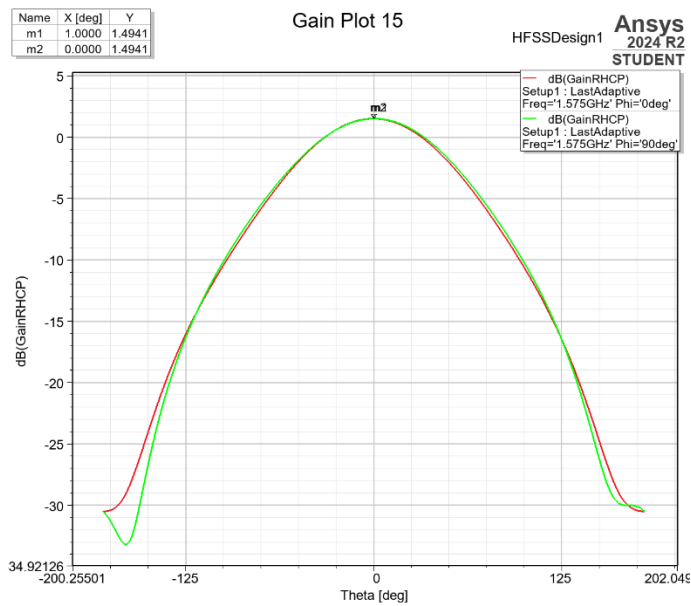
Gain Plot 7

Ansys
2024 R2
STUDENT



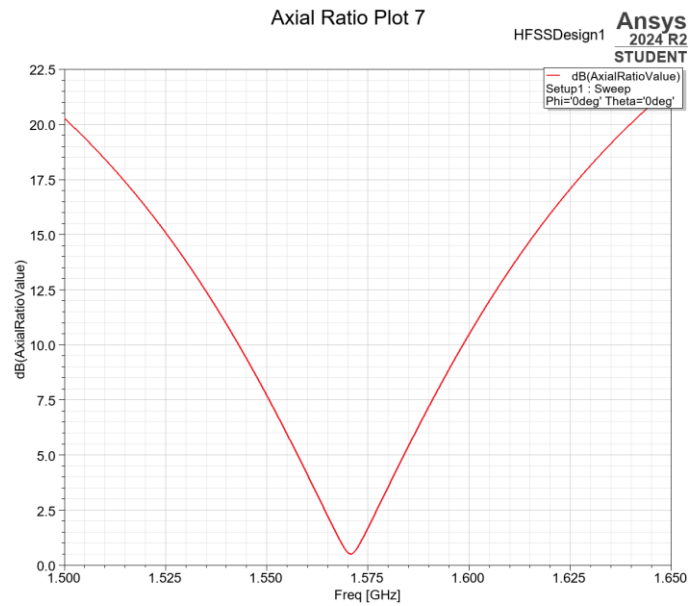
The radiation in the 3D plot is observed to be pretty spherical, although there is a small defect that can be explained with the axis ratio value (see two figures below).

Rectangular gain



The maximum gain is marked at 1.4941 dB, and as can be seen between -120 and 120 degrees, the traces for $\Phi = 0$ and $\Phi = 90$ are practically the same. Thus, this confirms the antenna has a RHCP configuration.

Axis Ratio



The axis ratio is shown to be well-below 3 dB (even 2dB) for the target frequency, giving the most circular possible polarization at it. Concretely, at 1.575 GHz the axis ratio value is around 2 dB (see figure above), but the lowest of all is 0.5 dB at around 1.570 GHz.

4 Conclusion

In this project, the following steps have been taken: first, the tutorial of HFSS simulating three different antennas has been carried out; after that, the choice of the final antenna has been made according to the research in the area; later, two models with different approaches have been designed for the specific requirements, although the results have not been achieved this way; finally, the approach presented in the tutorial has been retaken as a solution for the final results, and this report has explained each of theirs.

The general results have been the following:

- The classical approach had a similar geometrical polarization with a good axis ratio but a deceiving gain and return loss.
- The space-friendly approach had a great return loss and a good enough gain, but a more directional geometry that had little to do with RHCP.
- The square patch approach given in class satisfies all requirements.

In conclusion, the simplest of the approaches was in the end the one that worked out correctly. Generally, I believe that it is the duty of a student to combine what we study in class with new things that we learnt on our own to complete our formation, but in this case I overstepped myself by trying to implement things like a Wilkinson divider with a resistor or a 3D geometry with more than one substrate. I definitely underestimated the basics and I ended up designing complex architectures without completely understanding several concepts (my biggest challenge has been the coaxial feed line and assigning the excitation). Nevertheless, I have used new functions of HFSS for this purpose and, even if I did not achieve the more complex solutions, I can say I have a better overall understanding of the software that, together with a deeper review, will for sure be useful for my research project.

5 Acknowledgment

I would like to thank 王老师 and her assistants for the lectures and the support. It is my first course in antennas and I really enjoyed having an overall overview of the different types, being able now to identify them in my day to day or at least being able to have a technical conversation about them, which I think is essential for an electronic engineer like I want to become in the near future.

References

- [1] Y. Cao, S.W. Cheung, Li Liu, T.I. Yuk., "A Simple Planar Circularly Polarized Antenna for GPS System," DOI: [10.1109/TWAT.2014.6958685](https://doi.org/10.1109/TWAT.2014.6958685).
- [2] J. Yuan and Y. Li, "A Compact Circularly Polarized Microstrip Ring Antenna Using a Slotted Ground for GNSS Applications," Progress in Electromagnetics Research Letters, Vol. 88, pp. 29–36, 2020. DOI: [10.2528/PIERL20082707](https://doi.org/10.2528/PIERL20082707).